PORTFOLIO · FINTECH AI

SME Loan Pre-Screener

An explainable, fair AI co-pilot for SME working-capital assessment

Tshepiso Freddy Thosago · October 2026 · Synthetic data · Indicative only - human judgment is final

1. The problem

- SMEs are the heartbeat of the SA economy - and hit a brick wall seeking working capital.
- Credit processes are slow and opaque; good businesses wait weeks for an answer they cannot reason about.
- Credit officers drown in manual first-pass screening of clearly strong or clearly weak applications.

> **Vision:** not an automated decision-maker - an intelligent co-pilot that accelerates evaluation, surfaces critical insights, and keeps a human in charge of every decision.

2. The solution in one screen

Seven lightweight inputs (revenue, costs, debt, years trading, sector, loan amount, prior history) produce:

LIKELY TO QUALIFY **NEEDS REVIEW**
UNLIKELY TO QUALIFY

- A calibrated probability and confidence score
- Plain-language factors - why the score moved, up or down
- A prominent disclaimer: indicative only, human judgment final

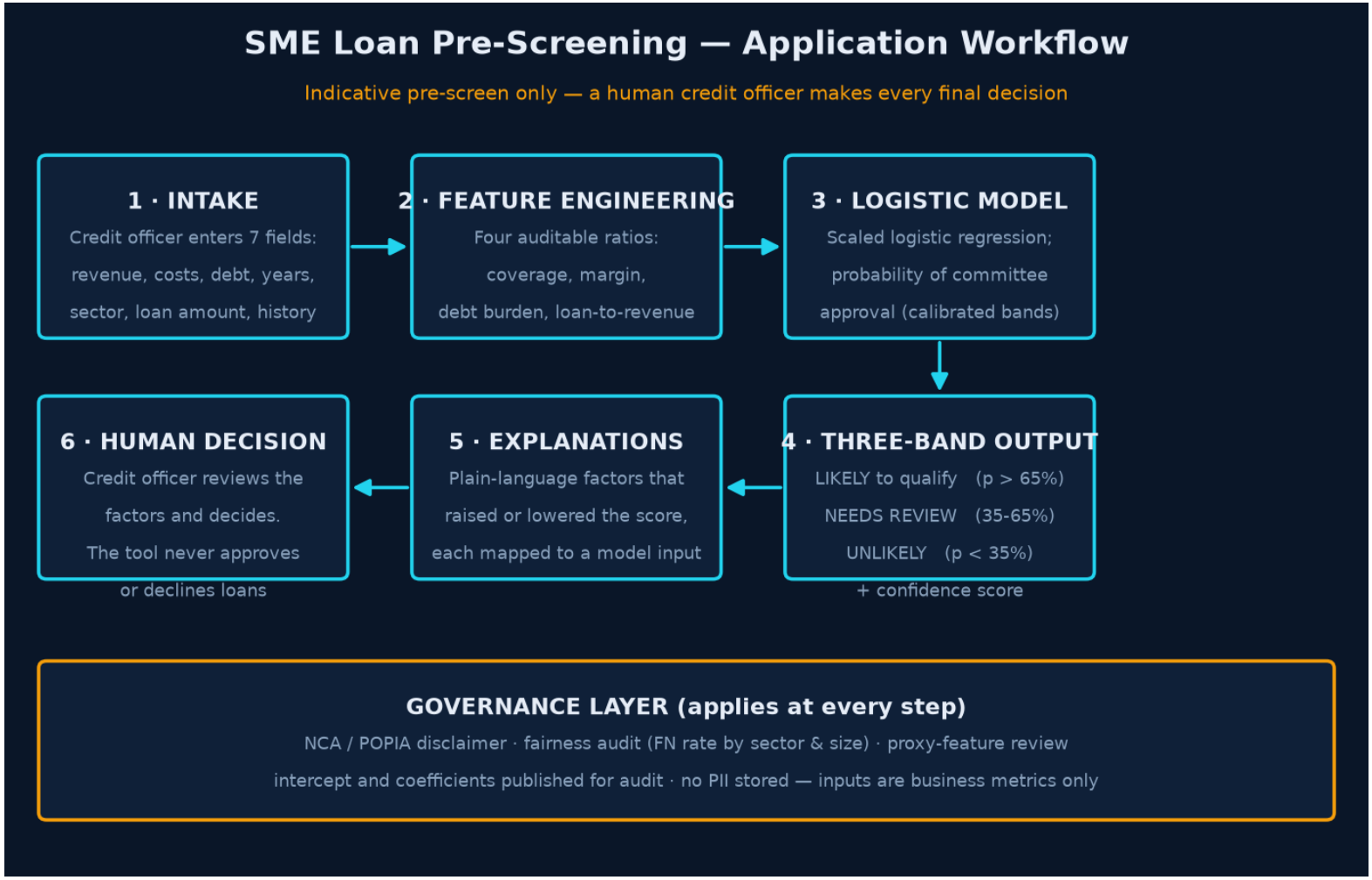
3 bands.

7 inputs.

0 black boxes.

Deployed model: logistic regression over affordability ratios - every coefficient is a readable lending rule.

3. Architecture - intake to outcome



4. Data foundation (synthetic, transparent)

- **800 seeded synthetic applications** - no real customer data anywhere.
- Labels engineered from a **simulated credit committee**: an affordability score (coverage, margin, 30% debt ceiling, trading years, history, sector premium) plus noise decides approval. Approval rate 33%.
- Debt-free applicants imputed to 0 explicitly - every data decision documented in code.
- Engineered ratios: instalment coverage, operating margin, debt burden, loan-to-revenue.

5. Models compared (held-out test set, n=200)

Model	Precision	Recall	F1	ROC-AUC
Rule-based baseline (30% ceiling, 2x coverage)	0.62	0.38	0.47	-
Logistic regression - deployed	**0.49**	**0.64**	**0.56**	**0.72**
Gradient boosting (comparison)	0.52	0.35	0.42	0.68

Why LR wins here: same-class discrimination as the booster, fully auditable, and the committee can challenge any factor. Recall is deliberately protected (balanced weights) - a false negative discourages a fundable SME; a false positive costs underwriting time.

6. Explainability - factors in credit-officer language

Example output for a risky application:

- ▼ Existing debt repayments consume 40% of monthly revenue, above the 30% affordability ceiling.
- ▼ Free cash of R0 does not cover the estimated new instalment of R20,833.
- ▼ The requested R600,000 is about 6.7 months of revenue - a large ask relative to turnover.
- ▲ Operating margin of 25% is above the 15% healthy-business bar.

> No unexplained scores: every displayed factor maps to a ratio the model actually consumed, and a human officer can challenge any of them in review.

7. Fairness audit - FN rate among approvable, by sector

Sector	N	FN rate
Construction	32	0.67
Manufacturing	23	0.50
Retail	49	0.42
Hospitality	26	0.25
Services	70	0.10

Disclosure: the construction/hospitality gap is engineered into the synthetic committee so the audit has a real disparity to surface. The production deliverable is the **monitoring pipeline**: monthly FN-rate gaps by sector and size, with persistent gaps triggering model review. Protected characteristics are excluded by design; revenue size is audited as a proxy-risk feature.

8. Responsible AI framing

- **Indicative only** - persistent banner on every interface; the tool never approves or declines.
- **NCA and POPIA** - production would require National Credit Act registration and POPIA compliance; declared out of scope.
- **Proxy features** - sector and business size retained for explainability but flagged and audited.
- **Simplicity bar** - a transparent, auditable model beats an uninterpretable one in financial services.

9. Live demo flow (5 minutes)

- Open **tshepisofrominnostation.github.io/sme-loan-prescreener** - enter a healthy retailer: **LIKELY** with coverage and margin factors.
- Raise debt repayments to 40% of revenue - watch the band flip and the debt-ceiling factor appear.
- Push years trading to 1 and loan to 6.7 months of revenue - **UNLIKELY**, factors explain exactly why.
- Find the borderline case that lands in **NEEDS REVIEW** - that is the human-judgment band working as designed.
- Full walkthrough script in docs/DEMO_SCRIPT.md; Python/JS probability parity verified to 4 decimals.

10. Limitations and future work

Known limits

- Synthetic committee labels approximate real credit behaviour
- Binary target, three bands recover the middle ground
- Prototype compliance posture, not a registered credit provider

Next

- Calibration curves and PR-AUC threshold tuning
- SHAP as a secondary explainer alongside ratio factors
- Lender-side A/B evaluation and drift monitoring

Repo: github.com/tshepisofrominnostation/sme-loan-prescreener · **Live:** tshepisofrominnostation.github.io/sme-loan-prescreener